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ABSTRACT

A characteristic feature of human blood rheology is a distinctive stress hysteresis during shear ramp up in the shear rate from zero, followed by a ramp back to zero. This is a result of the fact that human blood has a longer characteristic time of shear-induced rouleaux breakdown compared to the shear aggregation of the rouleaux. We demonstrate this telltale phenomenon of human blood rheology during the triangle ramp, time-dependent change in the shear rate. The unique hysteresis data are then used along with steady state data to fit parameters of a recently published thixo-elasto-viscoplastic rheological model, the tensorial enhanced structural stress thixotropic-viscoelastic (t-ESSTV) model. These best-fit parameter values from the hysteresis ramps are then used to predict step-up/down in shear, small amplitude oscillatory shear, uni-directional large amplitude oscillatory shear, and large amplitude oscillatory shear flow. Additionally, correlations between the calculated fitting parameters and physiological data are analyzed to inform the interpretation of model behavior in physical terms. The fit adherence to the triangle ramp and rheological hysteresis data are then evaluated alongside recently developed techniques to assess thixotropy via hysteresis loop area, indicating the efficacy of the t-ESSTV model in potentially predicting blood's complex characteristics in useful ways for future use in modeling blood flows under a variety of mechanical and biological conditions and predicting pathologies.

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I. INTRODUCTION

As a complex suspension, blood is primarily comprised of red blood cells (RBCs), white blood cells, and platelets within an aqueous plasma medium. Blood rheology is known to be correlated with poor health conditions such as cardiovascular diseases, stroke, or aging.[1–4] Since these correlations exist, researchers have suggested utilizing blood rheology in medical diagnostics.[5–7] In addition to this, characteristic times of structure (breakdown and buildup of rouleaux) can facilitate the characterization of blood including potentially diagnosing pathologies. Several researchers have correlated viscosity and yield stress to high cholesterol and triglycerides previously.[6,7] Due to the complexities presented by the many constituents and interactions in blood, the modeling of blood rheology is crucially important to decipher which aspects of rheology affect a disease state. Fully modeling the nuanced transient behavior has been a challenge for many years with most of the complexities arising at low shear rates.[7–10] When stagnant or at low shear conditions, the disk-like structure of blood's RBCs can couple with free-floating fibrinogen strands to agglomerate into "rouleaux" structures, characterized by the linear stacking of RBCs into microstructures reminiscent of coin stacks. The formation and disruption of rouleaux is reversible, varying with changes in blood stress and shear forces, as well as fibrinogen concentration.[6–12] Rouleaux evolution contributes to changes in blood viscosity, with the disintegration of rouleaux, seen in high shear regimes, leading to decreases in fluid viscosity. While not the sole contributor to blood's complex behaviors, rouleaux effects contribute to a portion of blood's non-Newtonian properties including viscoelasticity, viscoplasticity, and thixotropy.[9–12] Viscoelasticity refers to the presence of dual viscous and elastic characteristics in blood, while viscoplasticity suggests the presence of a non-zero yield stress threshold above which an

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otherwise elastic material undergoes irreversible deformation. Thixotropy describes the effects of microstructural re-arrangements on fluid viscosity over time and is often practically observed as shear-thinning behavior with unique timescales for viscous destruction and regeneration.[12–14] In addition to the formation and destruction of rouleaux under different flow conditions, the elastic deformation of individual RBCs contributes to blood's viscoelastic and thixotropic behaviors.[13–21] These complex characteristics and behaviors contribute to the demonstrated phenomenon known as hysteresis, specifically with regard to stress recovery over varying shear rate.

Shear stress hysteresis can be best understood in terms of a difference between a thixotropic materials' time constant of shear-induced microstructure breakdown, and shear-induced and Brownian microstructure aggregation. While hysteresis can be seen within by a wide variety of research areas such as electromagnetism and solid mechanics, rheological studies of complex fluid hysteresis have associated it primarily with microstructure-induced thixotropy.[22–25] Hysteretic behaviors are limited to those fluids prone to shear ranges close to the threshold of their system's thixotropic shear domain, which prevents full fluidization of the complex material.[26,27] A flow curve can be tailored to best demonstrate hysteresis through the experimental modulation of shear rate or stress.[28,29] Such experimentation may consist of step-up and step-down tests where applied shear forces are increased or decreased, respectively, from a default steady state and then held at a new steady-state value. These tests seldom show a truly comprehensive image if a system's hysteresis properties and thus requires a different protocol, starting and returning to zero shear rate. The resulting hysteresis loop from this cyclically imposed shear rate can vary with specific mechanical properties of the material and the total amount of time of the triangle ramp in the shear test, t_{max}.[30] This manifests in a varying relationship dissonance between shear stress, σ, and shear rate, $\dot{\gamma}$, on a σ-$\dot{\gamma}$ plot in the phases of ascending and descending shear rate.[31,32] To fully understand blood behavior, including hysteresis, the use of accurate and computationally effective models that incorporate the required viscoelastic and thixotropic characteristic times is necessary. This can simulate the more complex aspects of the blood's steady-state and transient rheology.[17,19]

As the general understanding of complex fluid rheology has evolved, so have its representative models. Initial generalized Newtonian models, such as the Casson, Carreau–Yasuda, and Herschel–Bulkley models, prove sufficient to represent most aspects of steady-state flow behavior, including shear thinning.[13,14,33–36] More sophisticated model frameworks are required to represent the nuances of transient behavior. Recent models, including that utilized in this effort, address transient complex behavior with the inclusion of a thixotropic component with three distinctive timescales: (1) one for rouleaux disruption under shear, (2) one for rouleaux formation under shear, and (3) one for Brownian aggregation (the tendency of the RBCs to form rouleaux under the appropriate stress conditions in our work shown to be related to the shear rate applied). Additionally, a kinetic rate equation is included featuring a nondimensional structure parameter varying between *0* and *1*, with *0* representing a complete lack of structure (and intercellular interactions) and *1* indicating complete structural formation (all "bonds").[37–44] The zero and one refer to the number of temporary (and total stress related) bonds to nearest neighbor red blood cells, holding the rouleaux together.

Recent developments of transient rheological models have investigated various approaches to accurately represent the complex behavior seen in blood. These efforts have spanned from modifying the Giesekus model[44,45] to the alteration of a general Oldroyd-8 model,[30] as seen in the work of Armstrong and Tussing[15] and Armstrong and Pincot,[44] to the use of the generalized White–Metzner Cross model in the development of the current-generation mHAWB model.[13,14] The mHAWB model, notable for its relative effectiveness in modeling transient behaviors, has seen further model development such as the ethixo-mHAWB approach.[15,46] Parallel efforts have focused on the development of the Herschel–Bulkley model with modifications conceived by Saramito, resulting in the ethixo elasto-viscoplastic (ethixo EVP) models, which allowed for thixo-elasto-viscoplastic (TEVP) descriptions of complex materials via the introduction of thixotropy via the non-dimensional structural parameter.[40,46,47] The Modified Delaware Thixotropic Model (MDTM) and its successor, the viscoelastic enhanced MDTM (VE-MDTM), were among the first models to clearly demonstrate the importance of including a unique viscoelastic timescale of stress contribution from rouleaux evolution to the total stress.[47,48] In addition to further exploration of applying MDTM to cast tensorial models by Pincot and Armstrong,[49] similar modifications to the SPTT-Isotropic Kinematic Hardening[20,41] TEVP model by Giannokostas *et al.*[50–52] would also include unique timescales for thixotropic and viscoelastic fluid response, shear structure breakage, and structure buildup. Computational rheology has also proven the feasibility of addressing these timescales via *in silico* flow protocols.[53]

The deliberate separation of these timescales maximizes the efficiency of complex models, allowing them to achieve high fitting and prediction capability while reducing the number of model parameters.[14,18,51] Despite its performance, however, recent efforts by Armstrong *et al.* have resulted in an evolution of the ethixo-mHAWB model to the Enhanced Thixotropic Viscoelastic (ETV) model.[54] The ETV model is essentially a streamlined ethixo-mHAWB model where the plastic component of the strain rate expression representing viscoelastic rouleaux contributions is replaced by the total strain rate. While this has insignificant effects on the predictive capability of the model, this modification facilitates further evolution of the model.

This work presents a more recently developed viscoelastic-elastic formulation expressing the elastic and viscoelastic stress contributions of rouleaux to produce the enhanced structural stress thixotropic viscoelastic (ESSTV) framework, building on plasticity theories.[40,41,54] The ESSTV models are then re-expressed in a full stress tensor description of the viscoelastic and total stress contributions of rouleaux microstructure, respectively, resulting in the t-ETV and t-ESSTV models. In an extension beyond these models' initial publication, this effort aims to apply the tensorial-ESSTV approach to assess shear-rate induced stress hysteresis in the blood. Triangle ramp tests were conducted to collect data on blood hysteresis behavior, allowing for the subsequent analysis of model capabilities via small, uni-directional, and large amplitude oscillatory shear (LAOS) analysis.[38] The fit of the model to observed data is gauged to fully understand the limitations and per-parameter efficiency of the model.

The rest of the manuscript is as follows: Sec. II is a brief presentation of materials and methods beginning with a description of the t-ESSTV rheological model; in Sec. III, we show the results of fitting the model parameters to the triangle ramp, hysteresis data; and finally, in Secs. IV and V, we present a discussion of the results and the conclusion, respectively.

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II. MODEL, MATERIALS, AND METHODS

A. Tensorial ESSTV rheological model background

Derived from the Horner-Armstrong-Wagner-Beris (HAWB) and modified HAWB (mHAWB) models, respectively,[13–15] Armstrong and Tussing's enhanced thixotropic (ethixo-mHAWB) model constitutes the basis of the improved tensorial ESSTV model.[52] However, the distinction between ETV and ESSTV is that ESSTV[53] integrates a term that allows the expression of elastic and viscoelastic stress contributions in a single term to encapsulate the model's prediction of elastoviscoplastic stress. The tensorial components of the t-ESSTV model, allowed for the improvement of the model efficiency parameter F_{cost}, calculated as the normalized difference between the data and model prediction; thus, the model is used here for the most accurate fits and predictions.[54] Armstrong *et al.* showed that by incorporating the tensorial version, there was an improvement in predictive capability of 15%.

B. Tensorial ESSTV rheological model description

The t-ESSTV (tensorial ESSTV) incorporates shear flow in a cartesian coordinate axis referencing x, y, and z, where x represents the flow direction, y the shear direction, and z the neutral direction. To represent the total shear stress σ_{xy}, ESSTV features two primary components, with $\sigma_{C,xy}$ representing viscoelastic individual red blood cell and plasma contributions and $\sigma_{R,xy}$ the individual rouleaux elastoviscoplastic contribution:[54]

$$\sigma_{yx} = \sigma_{R,yx} + \sigma_{yx,c}. \tag{1}$$

The viscoplastic response of the deformation of individual RBCs is first characterized by the following system featuring a White–Metzner model incorporating a Cross model to represent shear viscosity $\eta_c(\dot{\gamma})$:[13,14]

$$\eta_c(\dot{\gamma}) = \mu_{\infty,c} + \frac{\mu_0 - \mu_{\infty,c}}{(1 + \tau_C\dot{\gamma})} \tag{2}$$

and

$$\eta_c(\dot{\gamma})\dot{\gamma} = \sigma_c + \frac{\eta_c(\dot{\gamma})}{G_c}\sigma_c^{\nabla}, \tag{3}$$

where $\dot{\gamma}$ indicates the tensor of strain rate, G_c the elastic modulus, τ_C the Cross time constant, and μ_{c0} and $\mu_{c\infty}$, respectively, represent the zero and infinite viscosity of the RBCs. The Cross time constant is analogous to the time constant of viscoelastic evolution of the plasma and individual red blood cells. The nabla, ∇, present in Eq. (3), describes the following upper-convected time derivative:[54]

$$\overset{\nabla}{\sigma} \equiv \frac{\partial}{\partial t}\sigma + \mathrm{v}\cdot\nabla\sigma - \sigma\cdot\nabla\mathrm{v} - \nabla\mathrm{v}^T\cdot\sigma, \tag{4}$$

where v is the fluid velocity, and ∇v is the velocity gradient,

$$(\nabla\mathrm{v})_{ij} = \frac{\partial v_j}{\partial x_i},\ x_1 = x,\ x_2 = y, x_3 = z. \tag{5}$$

Equation (5) is recast in terms of stress components using the first invariant of the stress tensor to avoid model instabilities, applying it as the primary parameter of the viscosity function,[13,14,52,54]

$$\sigma_{yx,c} + \frac{\eta_c(\sigma_{xx,c})}{G_c}\frac{d\sigma_{yx,c}}{dt} = \eta_c(\sigma_{xx,c})\dot{\gamma} \tag{6}$$

and

$$\sigma_{xx,c} + \frac{\eta_c(\sigma_{xx,c})}{G_c}\left(\frac{d\sigma_{xx,c}}{dt} - 2\dot{\gamma}\sigma_{yx,c}\right) = 0, \tag{7}$$

where

$$\eta_c(\sigma_{xx,c}) \equiv \frac{-b + \sqrt{b^2 - 4c}}{2}. \tag{8}$$

It is considered that

$$b \equiv \tau_C\sqrt{\frac{G_c\sigma_{xx,c}}{2}} - \mu_{0,c} \tag{9}$$

and

$$c \equiv -\mu_{\infty,c}\tau_c\sqrt{\frac{G_c\sigma_{xx,c}}{2}}. \tag{10}$$

A thixotropic structure parameter, λ, ranging from *0* to *1* (as previously described) is introduced via previously established techniques to describe rouleaux microstructure evolution with *0* indicating a complete lack of structure and *1* suggesting complete formation (both of which describe the fraction of total bonds with nearest red blood cells).[12–15,44,48,51,54] It is described via the following relation of rouleaux formation to the blood shear rate:

$$\frac{D\lambda}{Dt} = \frac{1}{\tau_\lambda}\left(-\lambda\tau_{\mathrm{break}}|\dot{\gamma}_p| + (1-\lambda)|\tau_{\mathrm{agg}}\dot{\gamma}_p|^d + (1-\lambda)\right). \tag{11}$$

Equation (11) features three principal components, each describing a different aspect of the total thixotropy: (a) a shear breakdown term, $-\lambda\tau_{break}|\dot{\gamma}_p|$, with time constant of breakdown, τ_{break} and impacted by the shear rate and the structure parameter; (b) an aggregation term, $(1-\lambda)|\tau_{agg}\dot{\gamma}_p|^d$, with time constant, τ_{aggr}, proportional to the shear rate by a power of d and the structure parameter; and (c) a Brownian buildup term influenced by the aggregation term, $(1-\lambda)$. Literature suggests that d is equal to *1/2* for the purpose of blood analysis.[12–15,46,48,50,54]

Kinematic hardening is integrated into the ESSTV models through the separation of strain and shear rate into elastic and plastic modules, setting $\gamma = \gamma_e + \gamma_p$ and $\dot{\gamma} = \dot{\gamma}_e + \dot{\gamma}_p$, to describe the plastic strain, $\dot{\gamma}_p$, as

$$\dot{\gamma}_p = \begin{cases} \dfrac{\dot{\gamma}}{\left(2 - \dfrac{\gamma_e}{\gamma_{max}}\right)}, & \dot{\gamma} \geq 0, \\ \dfrac{\dot{\gamma}}{\left(2 + \dfrac{\gamma_e}{\gamma_{max}}\right)}, & \dot{\gamma} < 0, \end{cases} \tag{12}$$

where γ_e is the elastic strain.[12–15,46,48,50] The maximum rouleaux strain, γ_{max}, is further represented as

$$\gamma_{max} = \gamma_{0,R}\lambda, \tag{13}$$

where $\gamma_{0,R}$ represents the zero-shear-strain rate limiting elastic strain of the rouleaux microstructure. Further, the expression for the elastic strain rate is formulated as

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$$\dot{\gamma}_e = \begin{cases} \dot{\gamma}_p - \dfrac{\gamma_e}{\gamma_{max}}\left|\dot{\gamma}_p\right|, & \dfrac{D\dot{\gamma}_{max}}{Dt} \geq 0, \\ \dot{\gamma}_p - \dfrac{\gamma_e}{\gamma_{max}}\left|\dot{\gamma}_p\right| + \dfrac{\gamma_e}{\gamma_{max}}\dfrac{d\gamma_{max}}{dt}, & \dfrac{D\dot{\gamma}_{max}}{Dt} < 0, \end{cases} \tag{14}$$

which allows for the calculation of the steady-state structure parameter,

$$\lambda_{ss} = \frac{\left(\left|\tau_{agg}\dot{\gamma}\right|^d + 1\right)}{\left(\tau_{break}\left|\dot{\gamma}\right| + \left|\tau_{agg}\dot{\gamma}\right|^d + 1\right)}. \tag{15}$$

This steady-state structure parameter will range between zero and one, with one representing maximum possible rouleaux structure formation and zero indicating no interactions to nearby RBCs. Unlike the previous ETV model, the ESSTV[54] features a combined ethixo-viscoplastic module to represent rouleaux contributions to the total stress,

$$\frac{1}{G_R\lambda}\frac{D}{Dt}\sigma_{R,xy} + max\left(0, \frac{\left|\sigma_{R,xy} - \frac{\sigma_{y,0}}{\dot{\gamma}_{0,R}}\left|\dot{\gamma}_e\right|\right|}{\mu_R\lambda^m\left|\sigma_{R,xy}\right|}\right)\sigma_{R,xy} = \dot{\gamma}, \tag{16}$$

where G_R is the rouleaux elastic modulus, μ_R the rouleaux structural viscosity, and m a fitting parameter, suggested by standing literature to be set to *3/2*[12–15,46,48,50,54] The formulation for steady-state shear stress is identical to ESSTV's progenitor, ETV,[54]

$$\sigma_{ss} = \eta_c(\dot{\gamma})\dot{\gamma} + \mu_R\left(\lambda_{ss}^m\right)\dot{\gamma} + \sigma_{y,0}\lambda_{ss}. \tag{17}$$

Per procedures enumerated in the works of Armstrong *et al.*, Varchanis *et al.*, and Wei *et al.*,[41,54,55] ESSTV can be reformulated in a tensorial form through the exchange of the material time derivative, $\frac{D}{Dt}\sigma_{V,xy}$, in Eq. (16) with its upper-convected time analogues, $\sigma^{\nabla}_{V,xy}$ or $\sigma^{\nabla}_{R,xy}$. This acts to couple the shear module to other components, most importantly the normal stress.[54] In coupling the terms of the two tensor components and their time derivatives, dimensional frameworks are produced in the *xy*, *xx*, *yy*, and *zz* components. A general description of t-ESSTV can be found below in Table I, with the tensorial formulation of the model including the von Mises yield criteria incorporated here via the techniques described previously by Wei *et al.*[55] Here, in Table I, $\tilde{\sigma} = \sqrt{\frac{\underline{\underline{\sigma}}^D_R : \underline{\underline{\sigma}}^D_R}{2}}$, and $\underline{\underline{\sigma}}^D_R = \underline{\underline{\sigma}}_R - \frac{\mathrm{tr}(\underline{\underline{\sigma}}_R)}{3}\underline{\underline{I}}$, and $\tilde{\sigma}$ is the equivalent shear stress, *D* means deviatoric part of the tensor, $\underline{\underline{I}}$ is the unit tensor, ":" is the double dot product, and the trace operator is *tr(-)*.

C. Experimental protocol

The handling and rheological measurement protocol employed for the blood rheology in this work follows previous established guidelines for blood rheology and directly follows the protocol described by Horner and co-workers here.[12–15,18,54,56] A complete and detailed description of the blood draw procedure can be found here.[12–15,18,54,56] "For rheological experiments 6 ml of blood was drawn into a vacutainer tube with 1.8 mg/ml of EDTA. An additional 9 ml of blood was drawn and sent for a Complete Blood Count, Lipid Panel, and Fibrinogen Activity test."[12–15,18,54,56] The results are shown in Table II.[12–15,18,54,56] The double wall Couette geometry was chosen to minimize sedimentation over long periods of time. The experimental window to perform rheology experiments with human blood is approximately 4 h. During each step, the blood is kept at a constant shear rate for 40 s or a minimum strain of 40 due to slower shear rates requiring more time to approach the steady state stress. The rheological data from the donors can be found here.[58–62] All the data were collected within 4 h for best results, due to irreversible changes in the microstructure of blood after 4 h.[56]

We follow the work of Horner *et al.*:[13,14,56] "The rheological data were collected with an ARESG2 strain-controlled rheometer from TA Instruments with a double wall Couette."[12–15,18,54,56] The dimensions and measurable range can be found in previous studies by Horner *et al.*[12–15] "The tests were all performed at 37 °C, as maintained by a Peltier temperature controller. The shear rate did not exceed 1000 s^{-1}; a preshear of 300 s^{-1} for 30 s was incorporated before each test to eliminate memory effects.[12–15,18,53,55] The results are shown below for steady state, triangle ramp, step-up/down in shear rate tests, small amplitude oscillatory shear, small amplitude oscillatory shear (SAOS), amplitude sweep at $\omega = 10(\mathrm{rad/s})$, large amplitude oscillatory shear (LAOS), and unidirectional large amplitude oscillatory shear flow (UD-LAOS)."[12–15,18,54,56] The rheological data for donor 1 in Table II can be found on Mendeley Data.[53] The three-triangle ramp in shear experiments is defined via the shear rate as follows:

$$\dot{\gamma}(t) = \begin{cases} \alpha t & \mathrm{t} < \dfrac{t_{\max}}{2}, \\ \alpha(t_{\max} - t) & \mathrm{t} \geq \dfrac{t_{\max}}{2}, \end{cases} \tag{18}$$

where $\alpha(s^{-2})$ is a constant (acceleration for $t < \frac{t_{max}}{2}$; and deceleration for $t \geq \frac{t_{max}}{2}$), and t_{max} is the time of the triangle ramp in a shear experiment. The LAOS and UD-LAOS experiments presented in this work are identical to that described by Horner and co-workers[12–15,18,53,55] and Armstrong and co-workers[12–15,18,54,56] with strain and shear strain rates defined as

$$\gamma(\mathrm{t}) = \gamma_0 \sin(\omega \mathrm{t}) \tag{19}$$

and

$$\dot{\gamma}(\mathrm{t}) = \gamma_0 \omega \cos(\omega \mathrm{t}), \tag{20}$$

for the LAOS experiments and

$$\gamma(\mathrm{t}) = \gamma_0 \sin(\omega \mathrm{t}) + \mathrm{t}\gamma_0\omega, \tag{21}$$

$$\dot{\gamma}(\mathrm{t}) = \gamma_0 \omega \cos(\omega \mathrm{t}) + \gamma_0\omega, \tag{22}$$

for the UD-LAOS.[12–15,18,54,56] The UD-LAOS data and model predictions are plotted using effective Lissajous–Bowditch elastic and viscous projections at $\omega = 1\,(rad/s)$, where the measured stress is a function of the oscillatory component of the strain and shear rate as shown in Eqs. (19)–(22), respectively.[12–15,18,54,56]

The small amplitude oscillatory shear (SAOS) is in the standard form as follows:

$$\sigma_{yx}(\mathrm{t}) = \gamma_0 \sum_{\mathrm{i,odd}}^{\mathrm{n}} \left(\mathrm{G}'_\mathrm{i} \sin(\mathrm{n}\omega \mathrm{t}) + \mathrm{G}''_\mathrm{i} \cos(\mathrm{n}\omega \mathrm{t})\right), \tag{23}$$

where $n = 1$, for SAOS. Acknowledging that Eq. (23) is both a function of time *t*, frequency ω, and strain amplitude γ_0, we note that in LAOS, *n* is the running index (when not equal to one), and the upper limit of

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TABLE I. Summary of primary components of t-ESSTV. Reproduced with permission from Armstrong *et al.*, J. Rheol. **66**(2), 327–347 (2022). Copyright 2022 The Society of Rheology.

Model	Tensorial enhanced structural stress thixotropic viscoelastic
PAR.	$\mu_0, \mu_\infty, \tau_c, \sigma_{y0}, \tau_{break}, \tau_{aggr}, \gamma_{0,R}, G_c, G_r, \tau_\lambda, m, d$
SS	$\sigma_{ss} = \sigma_{ve,ss} + \mu_r \lambda_{ss}^m \dot{\gamma} + \sigma_{y,0}\lambda_{ss}\text{sign}(\dot{\gamma})$ $\sigma_{ve,ss} = \mu_\infty + \frac{(\mu_0 - \mu_\infty)}{(1+\tau_C\dot{\gamma})}$ $\lambda_{ss} = \frac{(\lvert\tau_{aggr}\dot{\gamma}\rvert^d + 1)}{(\tau_{break}\lvert\dot{\gamma}\rvert + \lvert\tau_{aggr}\dot{\gamma}\rvert^d + 1)}$ $\dot{\gamma}_p = \begin{cases} \frac{\dot{\gamma}}{\left(2 - \frac{\gamma_e}{\gamma_{max}}\right)}, \dot{\gamma} \geq 0 \\ \frac{\dot{\gamma}}{\left(2 + \frac{\gamma_e}{\gamma_{max}}\right)}, \dot{\gamma} < 0 \end{cases}$ $\dot{\lambda} = \frac{1}{\tau_\lambda}(-\lambda\tau_{break}\lvert\dot{\gamma}_p\rvert + (1-\lambda)\lvert\tau_{aggr}\dot{\gamma}_p\rvert^d + (1-\lambda))$ $\dot{\gamma}_e = \begin{cases} \dot{\gamma}_p - \frac{\gamma_e}{\gamma_{max}}\lvert\dot{\gamma}_p\rvert, & \frac{d\gamma_{max}}{dt} \geq 0 \\ \dot{\gamma}_p - \frac{\gamma_e}{\gamma_{max}}\lvert\dot{\gamma}_p\rvert + \frac{\gamma_e}{\gamma_{max}}\frac{d\gamma_{max}}{dt}, & \frac{d\gamma_{max}}{dt} < 0 \end{cases}$
VE, plasma	$\sigma_{yx,c} + \left(\frac{\eta_c(\sigma_{xx,c})}{G_c}\right)\frac{d\sigma_{yx,c}}{dt} = \eta_c(\sigma_{xx,c})\dot{\gamma}$ $\sigma_{xx,c} + \left(\frac{\eta_c(\sigma_{xx,c})}{G_c}\right)\left(\frac{d\sigma_{xx,c}}{dt} - 2\dot{\gamma}\sigma_{yx,c}\right) = 0$ $\gamma_{max} = \gamma_{0,R}\,\lambda \quad b = \tau_c\sqrt{\frac{G_c\tau_{xx,c}}{2}} - \mu_0$ $c = -\mu_{\infty,c}\tau_c\sqrt{\frac{G_c\sigma_{xx,c}}{2}}$ $\eta_c(\sigma_{xx,c}) = \frac{-b + \sqrt{b^2 - 4c}}{2}$
VE, rouleaux	$\frac{1}{\lambda G_R}\left(\frac{d\sigma_{R,xx}}{dt} - 2\sigma_{R,yx}\dot{\gamma}\right) + \max\left(0, \frac{\left\lvert\tilde{\sigma} - \left(\frac{\sigma_{y0}}{\gamma_{0,R}}\right)\gamma_e\right\rvert}{(\lambda^m\mu_R)\tilde{\sigma}}\right)\sigma_{R,xx} = 0$ $\frac{1}{\lambda G_R}\frac{d\sigma_{R,zz}}{dt} + \max\left(0, \frac{\left\lvert\tilde{\sigma} - \left(\frac{\sigma_{y0}}{\gamma_{0,R}}\right)\gamma_e\right\rvert}{(\lambda^m\mu_R)\tilde{\sigma}}\right)\sigma_{R,zz} = 0$ $\frac{1}{\lambda G_R}\frac{d\sigma_{R,yy}}{dt} + \max\left(0, \frac{\left\lvert\tilde{\sigma} - \left(\frac{\sigma_{y0}}{\gamma_{0,R}}\right)\gamma_e\right\rvert}{(\lambda^m\mu_R)\tilde{\sigma}}\right)\sigma_{R,yy} = 0$ $\frac{1}{\lambda G_R}\left(\frac{d\sigma_{R,yx}}{dt} - \sigma_{R,yy}\dot{\gamma}\right) + \max\left(0, \frac{\left\lvert\tilde{\sigma} - \left(\frac{\sigma_{y0}}{\gamma_{0,R}}\right)\gamma_e\right\rvert}{(\lambda^m\mu_R)\tilde{\sigma}}\right)\sigma_{R,yx} = \dot{\gamma}$
TRANS	$\sigma_{tot,i} = \sigma_{i,c} + \sigma_{i,r,ve}$

TABLE II. Relevant physiological parameter values of donors 1–5.[57]

Donor	Hematocrit (%)	Fibrinogen (mg/dL)	Total cholesterol (mg/ml)	Triglycerides (mg/ml)	HDL cholesterol (mg/ml)	LDL cholesterol (mg/ml)
1	42.2	0.246	148	29	56	85
2	43.4	0.248	147	40	51	85
3	38.3	0.333	145	144	39	82
4	43.8	0.199	197	82	78	102
5	43.1	0.249	244	94	70	154

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the summation goes to infinity. The ratio of the third to first harmonic is used as a metric of success of our model predictions as

$$\frac{I_3}{I_1} = \sqrt{\frac{(G_3'^2 + G_3''^2)}{(G_1'^2 + G_1''^2)}}, \tag{24}$$

where the subscript *3* indicates the third harmonic values.[12–15,18,54,56] Equation (24) is the relative intensity of the third harmonic normalized by the first harmonic intensity and is incorporated merely as a metric of model efficacy and shown on the SAOS prediction.[12–15,18,54,56]

D. Model fitting procedures

We follow a procedure, modified from the work of Horner *et al.* and Armstrong *et al.*[7–15,18,54,56,57] for fitting the full tensorial version of the ESSTV model. This required normalizing the cost functions for both the steady state, and three triangle ramp sets of data. The modification is as follows: (1) The initial parameter guess is provided from the best-fit values of the non-tensorial version, (2) the steady state and transient parameters are then fit simultaneously with the steady-state cost function accumulated with normalization of the stress measurement at each point, and (3) the three triangle ramps are normalized by maximum stress over the entire ramp, respectively. The cost function is, then,

$$F_{cost,tot} = \frac{1}{N_{ss}}\sqrt{\sum_{j=1}^{N_{ss}}\left(\frac{(y_j - f_j)}{y_j}\right)^2} + \frac{1}{M_{ramps}}\sum_{1=1}^{M}\frac{1}{K}\sqrt{\sum_{i=1}^{K}\left(\frac{(y_i - f_i)}{y_{K,max}}\right)^2}, \tag{25}$$

where N_{ss} is the total number of steady state data points, M_{ramps} is the total number of triangle ramps in shear rate tests, and K is the number of points per ramp experiment; $y_{i,j}$ is the stress measurement from the rheometer, and f_i is the model prediction.[12–15,18,54,56–58] The model is fit directly to rheology data from the ARESG2 using parallel tempering. Parallel tempering is chosen due to its robust design to find the global minimum while being insensitive to the initial guess. MATLAB's ode23s is incorporated to integrate the differential equations.[12–15,18,54,56–58] When fitting all the parameters together, $F_{cost,tot}$ must be non-dimensionalized to provide a universally relevant fit. With the parameters fit, they are kept *constant* to predict a series of: 3 step-ups and 3 step-downs in the shear rate; SAOS amplitude sweep at $\omega = 12.56$ (rad/s); four sets of LAOS at frequency 1 (rad/s), at strain amplitudes of 1.0, 5.0, 10.0, and 50.0 (-); and four sets of UD-LAOS experiments at a frequency of 1 (rad/s) and strain amplitudes of 1.0, 5.0, 10, and 50 (-). The predictions for each are shown below.[12–15,18,54,56,57]

III. RESULTS: MODEL FITTING AND PREDICTIONS WITH HYSTERESIS RHEOLOGICAL DATA

Figure 1 shows the results of the parameter fitting using the parallel tempering algorithm, with best-fit model parameters in Table III. For a TEVP model to fit the hallmark feature of thixotropic material indicates the t-ESSTV model's ability to capture all the dynamics of human blood through both steady-state and transient tests. The triangle ramp experiment itself is simply blood accelerating for a brief period ($t = \frac{t_{max}}{2}$), followed by a deceleration.

Figures 2(a) and 2(c) show in more detail two of the hysteresis fits shown in Fig. 1 for donor 3 but now plotted as stress vs shear rate.

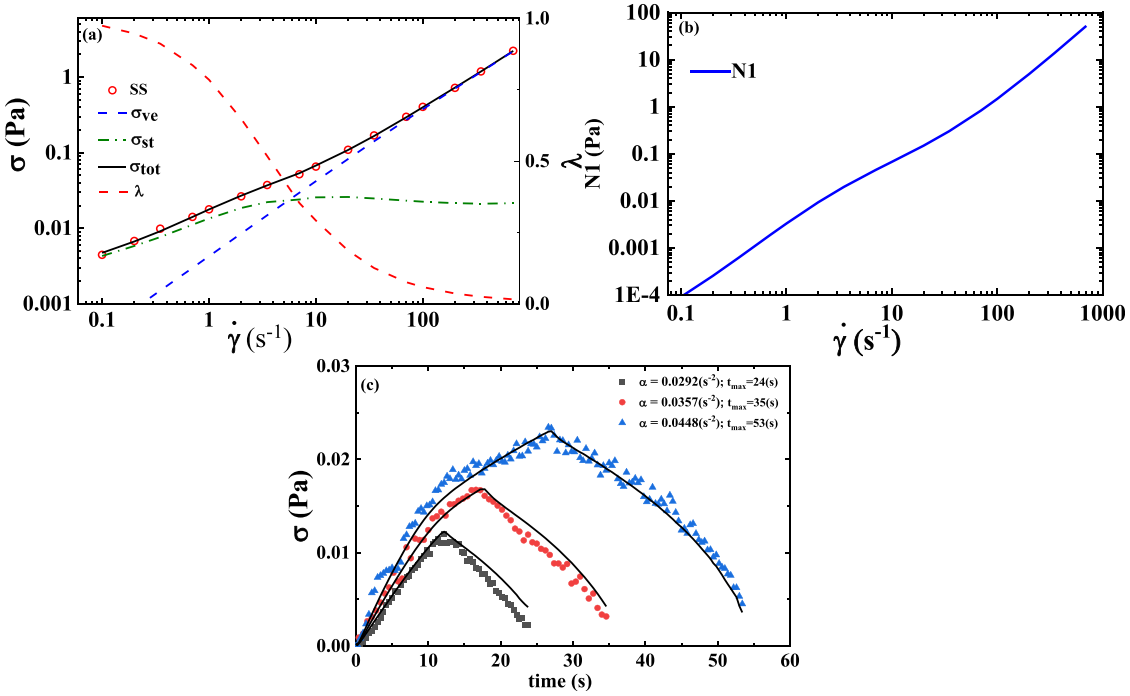


FIG. 1. t-ESSTV model fit to (a) steady state; (b) N1 predictions; and (c) a series of ascending triangle ramp stress tests simultaneously, with α and t_{max} values shown in figure; data represented by the discrete points, and solid black lines are model predictions. [Shear rate is calculated via Eq. (18).] (Donor3, dataset 3.)[60]

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TABLE III. Normalized hysteresis areas of triangle ramp experiments of $\alpha = 0.0357$ for donors 1–5 (donor 3 highlighted yellow). To facilitate comparison between samples, areas have been normalized by the experimental timescale of the donor 1 trials.

Donor	1	2	3	4	5
$\bar{A}_{\Delta\sigma}$ (-)	0.1217	0.2433	0.2236	0.2061	0.2100

One can follow the hysteresis using the numerals (1)–(4) depicted in Figs. 2(a) and 2(d), where (1) is the first leg of the hysteresis, followed by (2), where the hysteresis reaches the maximum shear rate at $t = \frac{t_{max}}{2}$, followed by (3), and (4), the return to zero shear rate. Notice that there are two different pathways followed due to the different time dynamics required to break down the coin-stack structures of rouleaux and then rebuild the rouleaux as the shear rate is decreasing. This is further elucidated in Figs. 2(b) and 2(d) where the numerals shown are the same place in the time evolution as that depicted in Figs. 2(a) and 2(c). At the beginning of the flow for both hysteresis loops shown in Fig. 2, one can see the rouleaux "loading" (aka. stretching apart, and breaking) with the stress, before finally yielding [shown near point 2 in Figs. 2(a) and 2(c)]. The rouleaux formation on the shear rate decline follows a different path in both cases. We note that the model is able to qualitatively show the hysteresis of human blood during the triangle ramp.[63,64]

Pioneering work of Divoux *et al.*,[65] which was later adapted by Jamali *et al.*[66] for colloidal gels, and more recently by Javadi and Jamali for blood,[53] provides a systematic method of calculating a characteristic thixotropic timescale from rheological hysteresis in a fluid. In practice, the area between the stress response of the fluid during ramps up and down protocols, referred to as the hysteresis area, can be calculated from the following:

$$A_{\Delta\sigma} = \int_{\dot{\gamma}_f}^{\dot{\gamma}_i} \sigma_{up} d\log\dot{\gamma} - \int_{\dot{\gamma}_f}^{\dot{\gamma}_i} \sigma_{down} d\log\dot{\gamma} = \int_{\dot{\gamma}_f}^{\dot{\gamma}_i} \Delta\sigma(\dot{\gamma}) d\log\dot{\gamma}. \quad (26)$$

Note that the integration is made using the logarithmic value of the deformation rate, to ensure that hysteretic effects at lower shear rates are not screened by small fluctuations at higher rates. Theoretically, the changes in the hysteresis area as a function of the wait time at each applied deformation rate can be directly calculated as a characteristic thixotropic timescale of the fluid.[66] On the one end of the spectrum, if t, describing the time taken to achieve max shear rate, is much shorter than the buildup timescale τ_{aggr}, then a relatively negligible hysteresis area may be observed due to the microstructure lacking the necessary time to buildup. On the other end, if t greatly exceeds the breakdown timescale, τ_{break}, the system naturally approaches a steady state and hysteretic effects vanish. Nonetheless, for measurements here, and in general with respect to experimental interrogation of

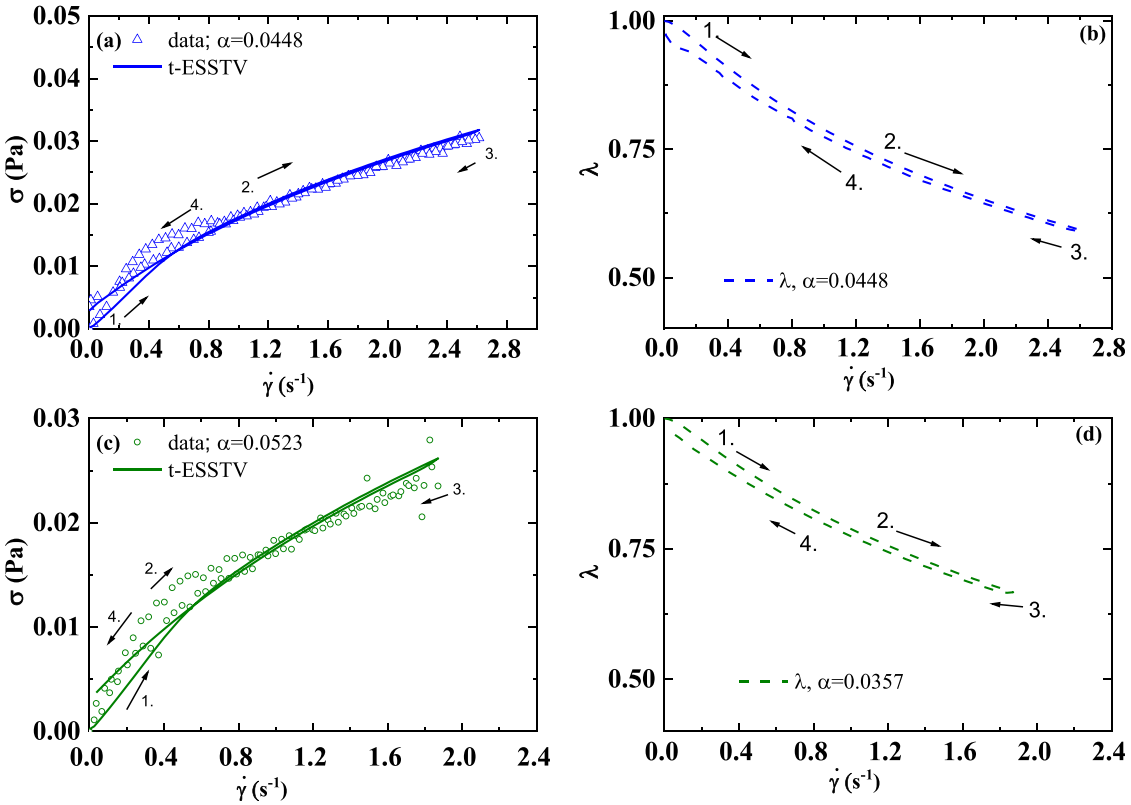


FIG. 2. t-ESSTV applied to descriptions of hysteresis at (blue lines and open triangles) $\alpha = 0.0448$ in descriptions of (a) stress–strain relationships and (b) structural evolution. Similar analyses of (c) stress–strain relationships and (d) structural evolution are subsequently seen for (green lines and open circles) $\alpha = 0.0357$. (Model predictions are solid lines; data are discrete points.) (Donor 3, dataset 3.)[60]

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TABLE IV. Normalized hysteresis areas of triangle ramp experiments of decreasing α values for donor 1. To facilitate comparison between samples, areas have been normalized by the experimental timescale of the $\alpha = 0.0497$ experiments.

Par.	$\alpha = 0.0497$	$\alpha = 0.049\,54$	$\alpha = 0.0448$	$\alpha = 0.0399$	$\alpha = 0.0375$
$\bar{A}_{\Delta\sigma}$ (-)	0.0132	0.0131	0.0112	0.0084	0.0079

blood rheology, it is virtually impossible to perform series of hysteresis measurements with different wait times on each applied shear rate. As such, here and keeping all other variables the same, a re-scaled hysteresis area is used as an analogue to the extent of thixotropy in each sample. In assessing the absolute value of this area, we focus on the magnitude of the thixotropic effects seen in the rheometric protocol. However, to provide a useful dimensionless metric, the up and down stress curves are normalized with σ_y and the shear rate values with τ_λ, producing the final dimensionless parameter defining the extent of thixotropy in the material,

$$\bar{A}_{\Delta\sigma} = \int_{\dot{\gamma}_f^*}^{\dot{\gamma}_i^*} \sigma_{up}^* d\log\dot{\gamma}^* - \int_{\dot{\gamma}_f^*}^{\dot{\gamma}_i^*} \sigma_{down}^* d\log\dot{\gamma}^*, \tag{27}$$

where $\dot{\gamma}_i^* = \dot{\gamma}_i\tau_\lambda$, $\dot{\gamma}_f^* = \dot{\gamma}_f\tau_\lambda$, $\sigma_{up}^* = \frac{\sigma_{up}}{\sigma_y}$, and $\sigma_{down}^* = \frac{\sigma_{down}}{\sigma_y}$. Best-fit values for normalizing parameters τ_λ and σ_y can be found in Table V. Normalized hysteresis values for the collected donor blood data can be seen in Tables III and IV. Table IV shows that the area values evolve with the evolution of the α values.

The best-fit model parameters are shown in Table V. Notice that the characteristic time of rouleaux breakdown, τ_{break}, is 1 order of magnitude greater than τ_{aggr}, thus describing how breakdown and buildup times are related, with the result of this shown in Fig. 2. In other words, once the rouleaux are broken down, relatively quickly, they take more time to re-form, in the presence of the applied stress from the rheometer. When the rouleaux are broken down during shear flow, typically, as seen in the triangle ramps LAOS and UD-LAOS featured in Fig. 5, there is not enough time to re-form the rouleaux. At slow shear rates, with a smaller applied stress, the rouleaux will start to re-form. Table V shows the averages, standard deviations, and 90% confidence interval of the model parameter values using our five donors. Note that donor 3 is used for representative descriptions for Figs. 1–5 and is highlighted in Table V.

Constitutive model parameters are typically correlated with physiology to be more useful to clinical diagnostics for either a healthy range of rheology or a prediction of physiology from rheology. Literature shows correlations of rheology to red blood cell volume fraction, fibrinogen concentration, and cholesterol.[43,67–71] These correlations make physical sense because they describe the volume occupied by solids and the levels of aggregation in the sample, which significantly impact a suspension's viscosity.[12] Finding a simple correlation among the parameters from this recent constitutive model and physiological parameters will discern the most relevant relationship. A Pearson correlation coefficient measures how well two parameters exhibit a linear correlation and in which direction it is. The p value for Pearson correlations determines statistical significance. These correlations serve as a comparison with correlations found in other works as well as potential suggestions for further research on quantitative connections of physiology and constitutive models.

Tables VI and VII show the correlation matrix of the best-fit rheological parameters and physiological parameters, as well as the p-value matrix, respectively. The hematocrit is the most correlated physiological parameter affecting the infinite shear viscosity and the rouleaux viscosity. Hematocrit is also positively correlated with the Cross time constant and time constant of shear-breakage of rouleaux. Fibrinogen possesses a markedly negative correlation with infinite shear viscosity, rouleaux viscosity, and the timescales for shear-induced structure build up and break down. The high-density lipoprotein cholesterol (considered "good" cholesterol) is positively correlated with infinite shear viscosity and rouleaux viscosity. Also notable is the negative correlation between the non-dimensional hysteresis area, as described by Eq. (27), with the microstructural timescale τ_λ. It is necessary to note here that due to the natural non-uniformity of blood between donor individuals in our experiments, the discussed correlations should be considered only useful guidelines rather than concrete conclusions.

TABLE V. t-ESSTV best-fit model parameters for donors 1–5 (Donor 3 highlighted yellow) with average, standard deviations, and 90% parameter confidence intervals.

Par.	Donor 1	Donor 2	Donor 3**	Donor 4	Donor 5	AVE	STDEV.	90% CI
μ_0 (Pa s)	0.008 01	0.005 64	0.004 27	0.005 18	0.007 67	**0.006 15**	**0.0014**	**0.0011**
μ_∞ (Pa s)	0.003 29	0.003 31	0.002 86	0.003 51	0.003 44	**0.003 28**	**0.0002**	**0.0002**
τ_C (s)	0.030 04	0.0288	0.0055	0.0291	0.0292	**0.024 54**	**0.0095**	**0.007**
τ_{break} (s)	1.163 21	1.1677	0.3054	1.1259	1.1559	**0.983 63**	**0.3394**	**0.2497**
τ_{aggr} (s)	0.042 60	0.2087	0.0080	0.2039	0.2090	**0.134 43**	**0.0898**	**0.066**
μ_R (Pa s)	0.029 56	0.0328	0.0158	0.0430	0.0351	**0.031 26**	**0.0089**	**0.0066**
σ_{y0} (Pa)	0.004 50	0.0034	0.0032	0.0031	0.0047	**0.003 79**	**0.0007**	**0.0005**
τ_λ (s)	1.934 62	0.6890	1.1273	1.4061	0.7558	**1.182 55**	**0.457**	**0.3362**
G_R(Pa)	0.235 02	0.1501	0.2326	0.2334	0.1024	**0.190 71**	**0.0547**	**0.0403**
G_c (Pa)	1.042 79	0.6234	0.3480	1.0431	0.6452	**0.740 51**	**0.2682**	**0.1973**
d	1/2	1/2	1/2	1/2	1/2			
m	3/2	3/2	3/2	3/2	3/2			
$\gamma_{0,R}$	1	1	1	1	1			

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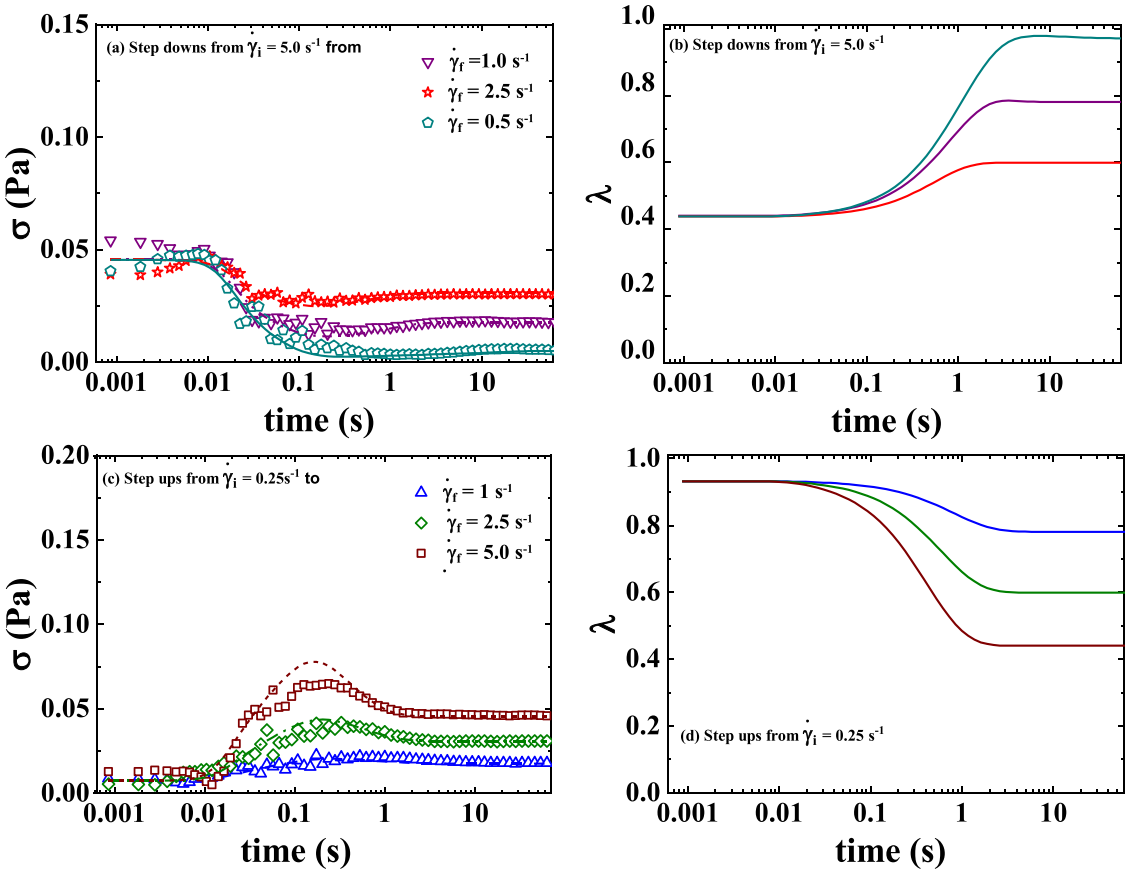


FIG. 3. t-ESSTV fit to a (a) series of step downs in shear rate, descending from $\dot{\gamma}_i = 5$ to 0.5, 1.0, and 2.5 s^{-1} with (b) accordingly colored continuous curves detailing the step down evolution of the structural parameter; (c) a series of step ups in shear rate, ascending from $\dot{\gamma}_i = 0.25$ to 1, 2.5, and to 5 s^{-1} with (d) accordingly colored continuous curves detailing the step-up evolution of the structural parameter (Donor 3, dataset 3).[60]

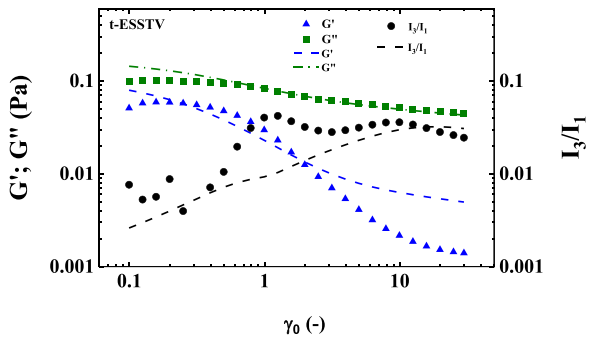


FIG. 4. t-ESSTV SAOS prediction of $\omega = 12.56$ (rad/s) amplitude sweep. Data are shown as discrete blue triangles (G') and green squares (G''), while the continuous blue dashed (G') and green dot-dash (G'') lines are model predictions (Donor 3, dataset 3).[60]

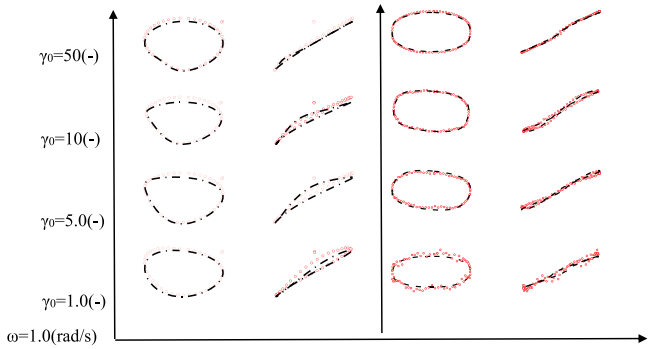


FIG. 5. t-ESSTV generation of (a) elastic and viscous UDLAOS projection components (perforated black lines) and (b) equivalent elastic and viscous LAOS projections (perforated black lines). Red dots indicate data, with the vertical axis representing strain amplitude (-) and the horizontal axis frequency (rad/s). Both analyses were prosecuted at $\omega = 1$ (rad/s), with a corresponding strain amplitude: $\gamma_0 = 1.0$, 5.0, 10.0, and 50.0. (Donor 3, dataset 3).[60]

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TABLE VI. Averaged donor correlation matrix of physiological and best-fit rheological model parameters. The most significant correlations are highlighted with red text. We note here that parameters need to be normally distributed to fully take advantage of these correlation coefficients, which is challenging to achieve with blood.

Corr. matrix	μ_0 (Pa s)	μ_∞ (Pa s)	τ_C (s)	τ_{break} (s)	τ_{aggr} (s)	μ_R (Pa s)	σ_{y0} (Pa)	τ_λ (s)	G_R(Pa)	G_c (Pa)
HCT (%)	0.4508	**0.9662**	**0.9538**	**0.9568**	0.8466	**0.9469**	0.22 748	−0.0882	−0.40 769	0.6666
fib (mg/dL)	−0.3650	**−0.9570**	**−0.8957**	**−0.8814**	−0.7141	**−0.9751**	−0.1121	−0.1776	0.108 939	−0.8413
Chol (mg/ml)	0.3668	0.6233	0.3894	0.3820	0.6187	0.5681	0.42 985	−0.3284	−0.6042	0.1329
TRI (mg/ml)	−0.5879	−0.5556	−0.8139	−0.8192	−0.2811	−0.4838	−0.337	−0.2942	0.079 378	−0.6626
HDL (mg/ml)	0.3434	**0.9124**	0.7114	0.6887	0.7011	**0.9216**	0.20 621	0.0744	−0.2131	0.6788
LDL (mg/ml)	0.4765	0.5228	0.3539	0.3548	0.5503	0.4233	0.58 215	−0.4110	−0.73 948	−0.0070
$\bar{A}_{\Delta\sigma}$ (-)	−0.6707	−0.1353	−0.3122	−0.2750	0.4593	−0.0292	−0.5749	**−0.8865**	−0.42 622	−0.6632

TABLE VII. Averaged donor p-values of physiological and best-fit rheological model parameters (less than 0.05 show significant correlation). Notable correlations are highlighted with red text.

P-values	μ_0 (Pa s)	μ_∞ (Pa s)	τ_C (s)	τ_{break} (s)	τ_{aggr} (s)	μ_R (Pa s)	σ_{y0} (Pa)	τ_λ (s)	G_R (Pa)	G_c (Pa)
HCT (%)	0.4461	**0.0074**	**0.0118**	**0.0107**	0.0704	**0.0146**	0.7129	0.8879	0.4957	0.2192
fib (mg/dL)	0.5458	**0.0106**	**0.0398**	**0.0482**	0.1754	**0.0047**	0.8576	0.7751	0.8616	0.0741
Chol (mg/ml)	0.5437	0.2613	0.5170	0.5257	0.2659	0.3177	0.4701	0.5895	0.2805	0.8313
TRI (mg/ml)	0.2972	0.3309	0.0937	0.0897	0.6468	0.4089	0.5792	0.6309	0.8990	0.2230
HDL (mg/ml)	0.5715	**0.0307**	0.1778	0.1985	0.1871	**0.0261**	0.7393	0.9053	0.7307	0.2077
LDL (mg/ml)	0.4172	0.3661	0.5590	0.5579	0.3365	0.4776	0.3031	0.4919	0.1532	0.9911
(-) $\bar{A}_{\Delta\sigma}$	0.2152	0.8283	0.6091	0.6543	0.4365	0.9628	0.3106	**0.0451**	0.4742	0.2224

The best-fit parameters are kept constant, and a series of increasingly challenging (to predict) rheological experiments are predicted: (1) three step-up and three step-down in shear data and predictions, Figs. 3(a) and 3(d); (2) small amplitude oscillatory shear flow, amplitude sweep data and predictions, Fig. 4; (3) uni-directional large amplitude oscillatory shear flow, Fig. 5(a); and (4) large amplitude oscillatory shear flow, Fig. 5(b). In all cases, the qualitative features are captured with the full set of model parameters simultaneously fit to steady state and three triangle ramps in shear rate experiments. With respect to the step-down in shear rate predictions in Figs. 3(a) and 3(b), the fits also suffice to adequately represent fit with experimentally derived data. The three step-ups in shear rate experiment predictions have peak stress predictions in the first 0.5(s) a bit above the analogous steady state stress value at same shear rate, showing overshoots at the larger of the three shear rates step-ups, but quickly fall back to the recorded (steady state) values of stress by approximately 1(s).

The SAOS and $\frac{I_3}{I_1}$ predictions shown in Fig. 4 again demonstrate a good fit with experimental data, but slightly over-predicted in the case of G'' at the lower strain amplitudes in the strain sweep. Our predicted G' and G'' values trend in the right direction, generally decreasing with increased strain amplitude. We correlate this effect with a decrease in the presence of the rouleaux as the strain amplitude is increasing which tends to do two things: (1) decrease G', the "solid-like" metric due to less rouleaux present; and (2) decrease G'', the "liquid-like" metric, also due to the decrease presence of rouleaux, as the rouleaux are also known contributors to the total stress through a viscous contribution.

The uni-directional large amplitude oscillatory shear (UDLAOS) and LAOS data and model predictions are shown in Figs. 5(a) and 5(b). These results are shown as a series of elastic and viscous projections, over a series of strain amplitudes ranging from 1.0–50 (-), at an oscillatory frequency of 1 (rad/s). In both sets of data and predictions, the model more quantitatively fits to the higher strain amplitudes better. The tensorial ESSTV model captures the qualitative features at all the strain amplitudes. This is a result of the rouleaux, which are present in a higher concentration at the UDLAOS and LAOS than at the lower strain amplitudes.[58,59] At the higher strain amplitudes, the rouleaux are almost completely broken down by not only the higher shear rates, but also the changing direction of the LAOS (the UDLAOS does not change direction, merely slows down, and speeds up in the same direction).

In both cases, UDLAOS and LAOS, we can see that the viscous projection shows that the blood (at these conditions of strain amplitude and oscillation) presents a more liquid-like signature approaching an almost straight diagonal line at the higher strain amplitudes. However, the elastic signatures validate that there is much less solid-like characteristics at all the strain amplitudes. This validates our understanding of human blood as a very minimally structured complex material, which exhibits its microstructure at lower shear rates only.

IV. DISCUSSION

The t-EESTV model, as seen in Fig. 1(a), adheres to previously recorded total stress data. Additionally, it is worthwhile to note trends of the increasing viscoelastic stress contribution compared to the decreasing structural parameter and structural stress contribution as the strain rate increases. The triangle ramp tests record similarly strong adherence to data for all three acceleration/deceleration constant tests, with a slightly poorer model fit during the deceleration stage, or step-down. The complex behavior seen in the ramp tests is in accord with

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prior experimentation with blood, where microstructure's contributions to the total viscoelastic stress tend to decrease with its disintegration in high shear environments.

Figures 2(a) and 2(c) both depict the hysteresis curves produced by analysis at $\alpha = 0.0448$ and at $\alpha = 0.0357$, respectively. Both curves demonstrate close adherence of the t-ESSTV model curve to the collected stress data, including the unique "loop" produced by stages 3 and 4 of the ramp analysis. The effect of blood's hysteretic properties on microstructure is observed in Figs. 2(b) and 2(d), which show decreased microstructure at high stress; however, of more interest is the presence of a similar hysteresis in microstructure adhesion. This suggests that a portion of blood's total hysteresis may stem from the aggregation history of its component rouleaux.

Further, in quantifying the magnitude of the hysteretic effects present in the ramp experiments across the various donor trials, Table III makes clear the relative similarity of the dimensionless hysteresis areas between the five donors. While variations do occur between the trials, the most substantial contribution to area differences is attributable to noisy data, as is the case with donor 1, and the unique blood physiology of each donor. However, subsequent observation of Table IV illustrates the more substantial differences in normalized hysteresis area that can be observed with changes in the α parameter. As shown in Fig. 1, greater alpha values correspond to larger t_{max} values as well as higher maximum stress values.

This corresponds to the increasing relevance of structural memory in the stress evolution of the blood during the step-down phase of the triangle ramp experiment. For smaller values of α, we can expect a reduced area due to the shorter experimental timescale preventing the full buildup of the microstructure. As α increases, we see, conversely, that the longer timescale allows a fuller evolution of the microstructure as reflected by greater hysteresis areas. Given a long enough timescale, reflecting even larger values of α, see even large hysteresis areas until a critical point at which the thixotropic behavior of the system reaches a steady state, thus closing the gap between the forward and reverse paths.

When assessing the importance of the hysteresis area, an extremely important correlation is revealed in Tables VI and VII; here, it is observed that the non-dimensional area is negatively correlated with the microstructural timescale τ_λ. Referring to the structure parameter evolution Eq. (11), which describes the microstructural basis of the thixotropy element of the t-ESSTV model, it is noted that increase in τ_λ minimizes the magnitude of the evolution parameter, while smaller τ_λ values act to heighten microstructural behavior. Thus, in observing the strong negative correlation between the normalized hysteresis area and τ_λ, it is reasonable to assert that assessing the hysteresis area is a useful metric to quantify relative thixotropic behavior between trials and, moreover, more concretely characterize a system's thixotropic properties. The relevance of all Table VI correlations is informed by the p-values present in Table VII, with p-values of less than 0.05 indicating statistically significant correlations. The relationships between the normalized hysteresis area values and the physiological parameters seen in Table II were also assessed as seen in supplementary material Tables S.I and S.II, but no statistically significant correlations were observed. While it was found surprising that there existed no obvious relationship between hematocrit and the area, this may be due both to the limited amount of physiological data and the natural variability of blood physiology.

Figures 3(a) and 3(b) reveal the behavior of blood under three step-down tests in terms of applied stress and the structural parameter. In these tests, t-ESSTV demonstrates high adherence to drawn data, with the structural parameter demonstrating expected inversely correlated behavior with stress. Figures 3(c) and 3(d) relate to three step-down tests, again relating applied stress and the structural parameter. Here, the model adheres well, though it somewhat overestimates the peak stresses encountered during the stepup, especially at high frequencies; though there exist several factors that could contribute to this behavior, this deviation could be due to having fitted the experimental parameters to data drawn from constant acceleration and deceleration experiments. This could lead to some instances of divergence between the model and data, as each step features a very short instance of acceleration or acceleration. Regardless, aside from the step response itself, t-ESSTV fits the average data well while maintaining relatively few fitting parameters, demonstrating a superior fit to data than other non-tensorial models such as ETV and the mHAWB lineage.

Figure 4 further demonstrates the accuracy of the model despite slight over prediction of G'' at low-medium shear rates. Prediction of G' may stray slightly from the shown data, but it is important to note that SAOS is hard to predict relative to other analysis techniques such as conventional stepup and down testing. As such, this performance is satisfactory when compared to peer models. Figures 5(a) and 5(b) reveal the results of UDLAOS and LAOS projections, respectively. The model is comprehensively effective at replicating the projections, especially at high shear rates. Characteristic of blood's unique complex properties, the model effectively portrays blood's increasingly viscous, or liquid-like, nature at high shear rates due to microstructure destruction. In summation, all analytical techniques in this effort confirm blood's nature as a TEVP material demonstrating hysteresis through evolving flow environments and, moreover, the efficacy of the recently developed t-ESSTV model at predicting its behavior.

V. CONCLUSION

We have shown that hysteresis is an important feature of the rheology of human blood, informing our understanding of human blood as a thixotropic material through the use of both classic and recently published analytical techniques. This confirms our understanding that human blood is a thixo-elasto-viscoplastic material as the structures undergo irreversible stretching (elasto-) and permanent stretching (plastic) during the rouleaux buildup and breakdown. The rouleaux in turn contribute to the overall stress with a viscoelastic time constant given by the rouleaux viscosity and rouleaux elastic modulus. Additionally, the plasma and individual RBCs contribute to the overall stress with their own unique viscoelastic time constant given by the plasma viscosity and elastic modulus.

The rheological behavior of human blood requires a model that can deliver on all the physics as outlined above, with parameters that directly tie to the observed phenomena. This detailed model is required so the parameters could be used to better understand the effect of diseases on rheology. This transient behavior must be fit using the rheological data from SAOS, LAOS, and UD-LAOS experiments. We have shown that the t-ESSTV model, when fit simultaneously to steady state and three sets of *triangle ramp* experiments, can do just that, extending upon previous explorations of SAOS, LAOS, and UD-LAOS predictions.[54] Previous work used step-up and step-down data; the hysteresis data have mechanical protocols embedded in a single experiment making it a more efficient experimental protocol. Furthermore, the model parameters can then predict the more challenging, transient rheology

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in SAOS, LAOS, and UDLAOS with higher relative efficiency to non-tensorial alternatives.

In examining the rheological characteristics of the hysteresis curves and their areas, we have shown that human blood is indeed thixotropic and that the t-ESSTV can not only be used to fit the hysteresis data to gain valuable mechanical information about human blood, but one can gain insight into the two viscoelastic characteristic times (one for plasma; one for the rouleaux) and insight into the relative speed of the shear-induced structure breakdown, and structure aggregation. In understanding these phenomena in blood and how to better capture and assess them using advanced TEVP modeling, future efforts in hematologic diseases and the design of microfluidic biomedical devices will be able to draw upon an existing suite of useful models and methods to inform treatment and design, respectively.

SUPPLEMENTARY MATERIAL

See the supplementary material for Tables S.I and S.II that contain cross correlation and p-value data for correlations between non-dimensional hysteresis area and averaged donor physiological parameters.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Andre Pincot: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Writing – original draft (equal). **F. John Burpo:** Formal analysis (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Elahe Javadi:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Writing – original draft (equal); Writing – review & editing (equal). **Safa Jamali:** Conceptualization (equal); Formal analysis (equal); Supervision (equal); Writing – review & editing (equal). **Sean Farrington:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Writing – original draft (equal); Writing – review & editing (equal). **Matthew J. Armstrong:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in Mendeley Data at https://doi.org/10.17632/s672yykhf2.1, Ref. 58, Mendeley Data at https://doi.org/10.17632/8pbktvbx2z, Ref. 59, Mendeley Data, https://doi.org/10.17632/948ffnypjs, Ref. 60, Mendeley Data, https://doi.org/10.17632/d9c6vk988t, Ref. 61, and Mendeley Data, https://doi.org/10.17632/kpjk7rctmw, Ref. 62.

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